Sardar Patel Institute of Technology

Bhavan's Campus, Munshi Nagar, Andheri (West), Mumbai-400058, India

(Autonomous College Affiliated to University of Mumbai)

End Semester Examination
Set - I

June 2025

Max. Marks: 100 | Duration: 3 Hrs
Class: TE CSE(DS/AIML) | Semester: VI
Course Code: DS306/AI306 | Program: B.Tech
Name of the Course: Distributed Computing

Instruction:

(1) All questions are compulsory
(2) Draw neat diagrams and write equations wherever necessary
(3) Assume suitable data if necessary

Course Outcomes:At the end of the course student will be able to:

(1) Understand the principles and desired properties of distributed systems. Apply the various communication techniques for distributed communication.
(2) Apply the concepts of process, naming, consistency, replication and faults tolerance in a distributed environment.
(3) Apply the algorithms such as clock synchronization, election, and mutual exclusion in distributed applications.
(4) Identify the challenges in developing distributed applications.

Q No.		Max. Marks	CO-BL
Q.1a	Consider a distributed system with 3 processes (P1, P2, P3) executing the following events: 1. P1 executes local event A 2. P1 sends message M1 to P2 (event B) 3. P2 receives M1 (event C) 4. P2 executes local event D 5. P2 sends message M2 to P3 (event E) 6. P3 receives M2 (event F) 7. P1 executes local event G 8. P1 sends message M3 to P3 (event H) 9. P3 receives M3 (event I) Initial Vector Clocks: P1 = [0,0,0], P2 = [0,0,0], P3 = [0,0,0] Tasks: 1. Compute the vector clock after each event. 2. Draw the event diagram. 3. Verify if event C $\rightarrow$ event F (happened-before relation).	10	CO3-BL3
Q.1b	Discuss some key concepts and techniques that designers of distributed operating systems can use to enhance system reliability? Additionally, what is the main challenge in achieving high reliability in distributed environments?	10	CO4-BL2
Q.2a	Scenario: A stock trading platform uses Remote Procedure Call (RPC) to fetch real-time stock prices from multiple market servers. Explain how RPC middleware enables this distributed communication and discuss its advantages and challenges.	10	CO1-BL3

Q2.b	Comment on Flynn's Taxonomy. Elaborate and discuss various parallel and distributed computer systems with the help of a diagram. OR Compare Distributed Operating Systems (DOS) and Network Operating Systems (NOS) in terms of architecture, transparency, and fault tolerance. Provide real-world examples of each.	10	CO1-BL2
Q.3a	In designing a distributed messaging system for a social media platform, what are the key reliability challenges that developers face in message delivery? Provide examples from platforms like Facebook or WhatsApp, and explain mechanisms to ensure reliable message delivery in the face of network failures or server crashes.	10	CO4-BL3
Q.3b	What is the role of acknowledgement messages in distributed computing protocols? Are acknowledgement messages always required for ensuring reliability in distributed systems? Justify your answer.	10	CO2-BL2
Q.4a	Consider a personal mailbox for a mobile user, implemented as part of a wide-area distributed database. What kind of client-centric consistency would be most appropriate? Describe a simple implementation of read-your-writes consistency for the same. OR It is often argued that weak consistency models impose an extra burden for programmers. To what extent is this statement actually true? What kind of consistency would you use to implement an electronic stock market? Explain your answer.	10	CO2-BL3
Q.4b	Scenario: A video streaming service (e.g., Netflix) experiences unpredictable traffic spikes during peak hours (e.g., weekend evenings). The service uses a cluster of servers distributed globally to deliver content. 1. Which load balancing technique (static or dynamic) is more suitable here? Justify. 2. Which load balancing strategy (Round Robin, Weighted Round Robin, or IP Hashing) would you recommend? Explain why. 3. What is a potential drawback of your chosen strategy?	10	CO3-BL4
Q.5a	A distributed system has 5 nodes (N1 to N5) with priorities: N5 > N4 > N3 > N2 > N1. At time t, N5 (current coordinator) fails. N2 detects the failure and initiates an election. 1. Simulate, with diagrams, the steps of the Bully Algorithm until a new coordinator is elected. 2. Calculate the total number of messages exchanged.	10	CO3-BL3
Q.5b	How do weak consistency and release consistency differ in a distributed system? In the context of designing a Distributed Shared Memory (DSM) system for an online multiplayer game, which consistency model would you prefer to use, and why?	10	CO2-BL3

Course outcome	**Marks in ESE**	Marks in MSE	Total marks for each CO
CO1	20	10	30
CO2	30	10	40
CO3	30	5	35
CO4	20	5	25